



6049 - Populating the Gap in Dust-Formation History of Type IIP Supernovae

Cycle: 3, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		MIRI Imaging	(1) SN-2003GD
	2		MIRI Imaging	(3) SN-2007IT

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	3		MIRI Imaging	(5) SN-2007OD
	4		MIRI Imaging	(7) SN-2012AW

ABSTRACT

Core-collapse supernovae, especially Type IIP SNe, have long been considered to significantly contribute to the cosmic dust budget. Models of expanding ejecta predict 0.1-1 solar masses of dust in SNe IIP. New dust cools quickly and is therefore detectable at longer (mid-IR) wavelengths. However, only a handful of nearby SNe IIP have shown direct observational evidence for dust condensation (based primarily on Spitzer data), and dust masses (generally limited to early epochs, and to >500 K temperatures) have been 2-3 orders of magnitude smaller than theoretical predictions. At the same time, more recent observations of Galactic SN remnants (SNRs) and the very nearby (~50 kpc) SN 1987A have revealed that massive cooler dust reservoirs may form over years or decades in SNe. However, no other SNe II(P) have significant amounts of newly-formed ejecta dust measured < ~5 yrs after explosion up till now.

As demonstrated by recent results on two nearby SNe IIP, JWST finally allows hidden cool (~100--200 K) dust reservoirs in a larger sample of extragalactic SNe to be revealed and to help fill the gap currently existing in the dust-formation history for SNe.

Here we propose 7-filter JWST MIRI Imaging of 4 of the most dusty, nearby SNe IIP that will be age > 5 yr by the time of expected Cycle 3 observations. Via modeling their expected mid-IR spectral energy distributions, we will estimate or place stringent constraints on the dust mass for each SN. The deeper understanding of dust mass evolution would also would reveal how dust opacity evolves with time and would provide a more appropriate characterization of the composition of dust grains.

OBSERVING DESCRIPTION

This proposal will image 4 nearby SNe IIP with MIRI Imager. Each observation will consist of F1000W, F1130W, F1280W, F1500W, F1800W, and F2100W filters. We also request F560W imaging. Even though we don't expect to necessarily detect the SN at these wavelengths, we request the data for image and source alignment with archival Spitzer 4.5 micron and HST near-IR imaging of the SN when it was younger to ensure we are able to identify our source in the other filters. Note that SN 2003gd has been recently detected by JWST/NIRCam, while, on SN 2007it, there are obtained MIRI F2100W observations (thus, in the case of this one object, we do not include F2100W filter imaging in our program.)

Proposal 6049 - Targets - Populating the Gap in Dust-Formation History of Type IIP Supernovae

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SN-2003GD	RA: 01 36 42.6500 (24.1777083d) Dec: +15 44 19.90 (15.73886d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Supernovae, Type II supernovae]				
	(3)	SN-2007IT	RA: 14 18 25.6300 (214.6067917d) Dec: -43 22 53.80 (-43.38161d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Supernovae, Type II supernovae]				
	(5)	SN-2007OD	RA: 23 55 48.6800 (358.9528333d) Dec: +18 24 54.80 (18.41522d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Supernovae, Type II supernovae]				
	(7)	SN-2012AW	RA: 10 43 53.7350 (160.9738958d) Dec: +11 40 17.63 (11.67156d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Supernovae, Type II supernovae]				

Proposal 6049 - Observation 1 - Populating the Gap in Dust-Formation History of Type IIP Supernovae

Observation	Proposal 6049, Observation 1										Fri Mar 01 05:04:14 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous		
	(1)	SN-2003GD	RA: 01 36 42.6500 (24.1777083d)				Epoch of Position: 2000				
			Dec: +15 44 19.90 (15.73886d)								
			Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Star										
	Description=[Supernovae, Type II supernovae]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	15	1	1	Dither 1	4	4	166.502	
	2	F1000W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	3	F1130W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	4	F1280W	FASTR1	10	1	1	Dither 1	4	4	111.002	
	5	F1500W	FASTR1	10	1	1	Dither 1	4	4	111.002	
	6	F1800W	FASTR1	10	1	1	Dither 1	4	4	111.002	
	7	F2100W	FASTR1	10	4	1	Dither 1	4	16	477.307	

Proposal 6049 - Observation 2 - Populating the Gap in Dust-Formation History of Type IIP Supernovae

Observation	Proposal 6049, Observation 2										Fri Mar 01 05:04:14 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	SN-2007IT	RA: 14 18 25.6300 (214.6067917d) Dec: -43 22 53.80 (-43.38161d) Equinox: J2000			Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Supernovae, Type II supernovae]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	4	F1280W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	5	F1500W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	6	F1800W	FASTR1	20	1	1	Dither 1	4	4	222.003	

Proposal 6049 - Observation 3 - Populating the Gap in Dust-Formation History of Type IIP Supernovae

Observation	Proposal 6049, Observation 3										Fri Mar 01 05:04:14 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	SN-2007OD	RA: 23 55 48.6800 (358.9528333d)			Epoch of Position: 2015.5					
			Dec: +18 24 54.80 (18.41522d)								
			Equinox: J2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Star										
	Description=[Supernovae, Type II supernovae]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	2	F1000W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	3	F1130W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	4	F1280W	FASTR1	60	1	1	Dither 1	4	4	666.01	
	5	F1500W	FASTR1	60	1	1	Dither 1	4	4	666.01	
	6	F1800W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	7	F2100W	FASTR1	14	8	1	Dither 1	4	32	1320.919	

Proposal 6049 - Observation 4 - Populating the Gap in Dust-Formation History of Type IIP Supernovae

Observation	Proposal 6049, Observation 4										Fri Mar 01 05:04:14 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	SN-2012AW	RA: 10 43 53.7350 (160.9738958d) Dec: +11 40 17.63 (11.67156d) Equinox: J2000			Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Supernovae, Type II supernovae]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	10	1	1	Dither 1	4	4	111.002	
	2	F1000W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	3	F1130W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	4	F1280W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	5	F1500W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	6	F1800W	FASTR1	20	1	1	Dither 1	4	4	222.003	
	7	F2100W	FASTR1	10	4	1	Dither 1	4	16	477.307	